

Question Paper

Exam Date & Time: 24-Dec-2020 (10:00 AM - 01:45 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fifth Semester of B. Tech (IT) Examination

Design and Analysis of Algorithms [IT341]

Marks: 70

Duration: 225 mins.

Section : I-A (MCQ)

Answer all the questions.

- 1) Which one of the following is the recurrence equation for the worst case time complexity of the Quick sort algorithm for sorting n ($n \geq 2$) numbers? Consider c is a constant. (1)
[T\(n\) = 2T\(n/2\) + cn](#) [T\(n\) = T\(n-1\) + T\(0\) + cn](#) [T\(n\) = 2T\(n-2\) + cn](#) [T\(n\) = T\(n/2\) + cn](#)
- 2) Assume that a merge sort algorithm in the worst case takes 30 seconds for an input of size 64. Which of the following most closely approximates the maximum input size of a problem that can be solved in 6 minutes? (2)
[256](#) [512](#) [1024](#) [2048](#)
- 3) Consider F1, F2, F3 and F4 are different algorithms to solve a particular problem with the order of $\log(n)$, $n\log(n)$, $\log(\log(n))$, and $n/\log(n)$ respectively. Which is the best algorithm? (1)
[F1](#) [F2](#) [F3](#) [F4](#)
- 4) The worst case complexity of Binary Search Algorithm is _____. (1)
[N](#) [N^2](#) [Nlog\(N\)](#) [log\(N\)](#)
- 5) What is the time complexity of the given code in terms of Big Oh notation? (2)

```
void F(int A[], int n)
```

```
{
```

```
int i = 0, j = 0;
```

```
for(; i < n; ++i)
```

```
while(j < n && A[i] < A[j])
```

```
j++;
```

```
}
```

[O\(n\(log n\)^2\)](#)

[O\(nlog n\)](#)

[O\(n^2\)](#)

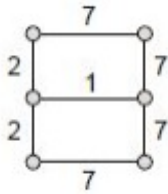
[O\(n\)](#)

- 6) The time factor when determining the efficiency of algorithm is measured by _____. (1)
[Counting microseconds](#) [Counting the number of statements](#) [Counting the number of key operations](#) [Counting the kilobytes of algorithm](#)
- 7) What is the running time complexity of Binary Knapsack problem using dynamic programming? Consider N is the number of items and W is the capacity of Knapsack. (1)
[O\(N^2\)](#) [O\(N^W\)](#) [O\(W^N\)](#) [O\(NW\)](#)
- 8) _____ is not a dynamic programming based algorithm. (1)
[Floyd Warshall Algorithm](#) [Bellman-Ford Algorithm](#) [Prim's Minimum Spanning Tree Algorithm](#) [0-1 Knapsack problem](#)

- 9) For the Spanning Tree, select correct option.: (a) A spanning tree is a connected graph (b) Number of edges in a spanning tree with n -vertices is $n - 1$ (c) Spanning tree contains cycles (d) Spanning tree has directed edges. (1)

[only a and c](#) [only a and b](#) [only c and d](#) [only b and d](#)

- 10) For the given graph apply Kruskal's algorithm, how many distinct minimum cost spanning trees are possible? (2)



[3](#) [4](#) [5](#) [6](#)

- 11) Consider the following function from positive integers to real numbers: (1)

$10, n^{0.5}, n, \log n, 100/n$

Find the correct arrangement of the above functions in increasing order of asymptotic notation complexity is:

[log n, 100/n, 10, n^0.5, n](#) [100/n, 10, log n, n^0.5, n](#) [10, 100/n, n^0.5, log n, n](#) [100/n, log n, 10, n^0.5, n](#)

- 12) Class NP stands _____. (1)

[Non Polynomial](#) [Non Probabilistic](#) [Non Decidable Polynomial](#) [Non Deterministic Polynomial](#)

- 13) Branch and Bound approach is used to solve Optimization Problem (1)

[True](#) [False](#)

- 14) NP-Complete $\cap$ P = _____ which of the following is true? (1)

[NP-Hard](#) [P](#) [NP-Complete](#)

- 15) Topological ordering is not applied to cyclic graph. (1)

[True](#) [False](#)

- 16) What is the time complexity of the given code? (2)

```
for(i=2; i<=m; )
```

```
{
```

```
  i=i3;
```

```
}
```

[O\(m\(log m\)^2\)](#) [O\(log₂\(log₃m\)\)](#) [O\(log₃\(log₂m\)\)](#) [O\(m^3\)](#)

Section : I -B (Descriptive)

Answer all the questions.

- Answers must be handwritten.
- Upload answers in portrait mode only.
- Answers must be in readable format.

- 1) Attempt Following. (4)

1. Why Greedy algorithms do not provide globally optimized solutions for all the problems?
2. Consider that you have infinite number of coins for the given coin set {1, 5, 8, 25, 50, 100}. You have to give N rupees to your friend with minimum number of coins. Find out two

different amount in the range of (1,100} for which the greedy algorithm do not provide optimal solution. Provide set of denominations for each amount.

2) Attempt Following. (4)

1. Why is the worst case analysis of algorithms more important than best case analysis?
2. Find out complexity of the given code with justification.

```
void function(int n)
{
    int count = 0;
    for (int i=n/2; i<=n; i++)
        for (int j=1; j<=n; j = 2 * j)
            for (int k=1; k<=n; k = k * 2)
                count++;
}
```

3) Using Backtracking approach: (7)

- a. Show the first solution of 6 - Queen Problem using prune state space tree.
- b. What are the implicit and explicit constraints for the 6 - Queen Problem?

4) Which design strategy is used by Merge sort algorithm? Explain performance of the merge sort algorithm using recurrence equation. (3)

[OR] Which design strategy is used by Binary Search algorithm? Explain performance of the Binary Search algorithm using recurrence equation. (3)

5) Derive the time complexity of Quick Sort algorithm for worst case? How can we improve worst case behavior of Quick Sort algorithm? Explain with suitable example. (7)

[OR] Attempt Following (7)

1. Identify and apply appropriate design technique to solve given knapsack problem. The capacity of the knapsack is 50 Kg. Consider the given six Items with its Value and Weight. You have to satisfy the given capacity of the knapsack with maximum value.

Item	1	2	3	4	5	6
Value	125	100	120	95	135	156
Weight	10	8	5	12	20	11

2. Compare Prim's and Kruskal algorithm for Minimum Spanning Tree.

Section : II-A (Descriptive)

Answer all the questions.

- Answers must be handwritten.
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- Answers must be in readable format.

1) Solve following recurrence using Master Theorem. (4)

1. $T(n) = 8T(n/2) + n^2$
2. $T(n) = 8T(n/2) + n^3 \log n$

- 2) State True or False. An asymptotically fast algorithm running on slow computer is better than asymptotically slow algorithm is running on fast computer. Justify your answer. (2)
- 3) Find the longest common subsequence for the following two sequences using dynamic programming. Show the complete process. (4)

X = 100101001

Y = 101001

- [OR]
4) Find the binomial coefficient $C(10,8)$ using dynamic programming. (4)

Section : II-B (Descriptive)

Answer 3 out of 4 questions.

- Answers must be handwritten.
- Upload answers in portrait mode only.
- Answers must be in readable format.

- 1) Consider given code and generate recurrence relation. (5)

```
void Test (int n)
{
if (n>0)
{
for (i=0 ; i< n ; i++)
{
printf ("%d", n);
}
Test(N-1);
}
}
```

Provide solution of recurrence using tree method and back substitution method.

- 2) Following four matrices are given with corresponding dimensions. (5)

A= 40 x 20, B= 20 x 30, C= 30 x 10, D= 10 x 30

Find the order of their multiplication such that total number of multiplications should be minimized.

- 3) What is String Matching? Discuss Naïve String Matching algorithm with appropriate example. (5)
- 4) Define P, NP, NP-Complete and NP-Hard problems. Draw a relationship diagram between P, NP, NP-Complete and NP-Hard. (5)

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